

# Unit-3

## Network Fundamentals and Security in IT Infrastructure Services

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# Unit 3: Syllabus

- Basics of Networking Protocols
- Fundamentals of network reference models (TCP/IP, OSI)
- IP addressing
- Network Topologies and Models; Fundamentals of DNS, Web and File Server
- Principles of Information Security
- Common Security Threats and Vulnerabilities
- Security Best Practices for Servers and Networks

# Networking Protocol

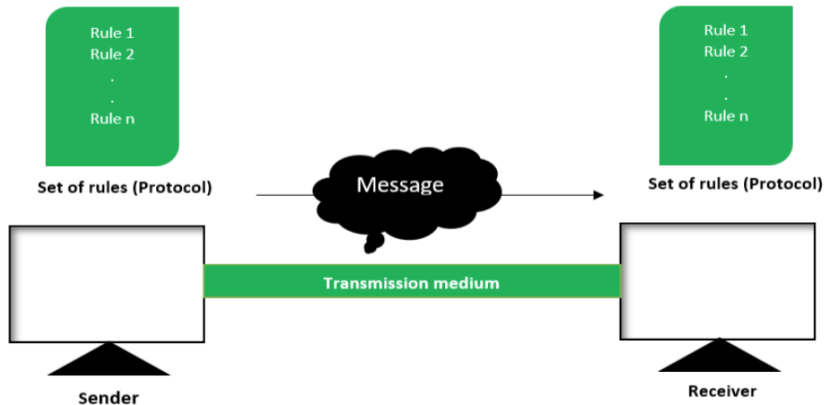
A networking protocol is a set of rules that allows computers and devices to communicate with each other over a network or the internet.

Examples:

- Sending email
- Browsing websites
- Watching YouTube
- Transferring files
- Video calling

All these use different protocols.

# Networking Protocol



# Networking Protocol

## HTTP (HyperText Transfer Protocol)

- Application layer protocol used for web communication
- Enables communication between browser and web server
- Used to access and load web pages
- Follows request–response model
- Transfers text, images, videos, and web content
- Stateless protocol
- Common methods: GET, POST, PUT, DELETE
- Uses Port 80 (TCP)

**Example:** Opening a website in Google Chrome or Firefox.

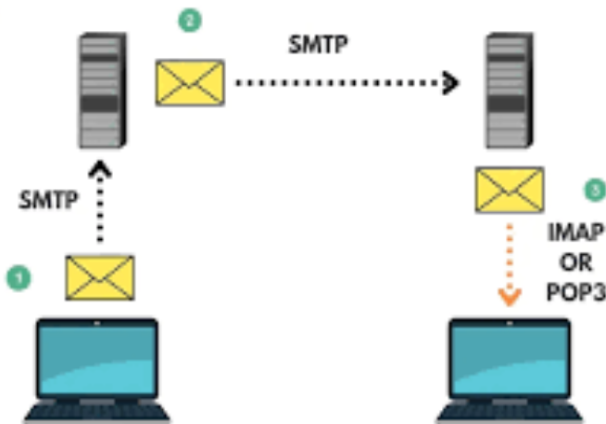
# Networking Protocol

## FTP (File Transfer Protocol)

- Used for transferring files between computers
- Allows uploading and downloading files over a network
- Works on client-server architecture
- Commonly used for website management and backups
- Supports file sharing and remote access
- Uses username and password for authentication
- Runs on Port 21

**Example:** Uploading website files from a computer to a web server.

# Networking Protocol(SMTP/POP/IMAP)



# Networking Protocol

## SMTP (Simple Mail Transfer Protocol)

- Protocol used for sending emails over the internet
- Transfers emails from sender to mail server
- Works on client-server architecture
- Used with POP3 or IMAP for complete email communication
- Ensures reliable delivery of outgoing emails
- Commonly used by email services like Gmail and Outlook
- Runs on Port 25

**Example:** Sending an email from Gmail to another user.

# Networking Protocol

POP3 (Post Office Protocol Version 3) and IMAP (Internet Message Access Protocol)

- Protocols used for receiving emails from mail servers
- POP3 downloads emails to a single device
- IMAP synchronizes emails across multiple devices
- IMAP keeps emails stored on the server
- POP3 can access emails offline after download
- IMAP is better for mobile and multi-device access
- Commonly used with email applications like Gmail and Outlook

**Example:** Checking emails on mobile phone and laptop using Gmail.

# Networking Protocol

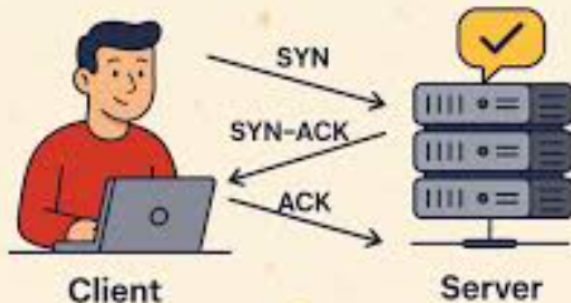
## DNS (Domain Name System)

- Protocol that converts domain names into IP addresses
- Helps users access websites using easy names instead of numbers
- Acts like the internet's phonebook
- Makes web browsing faster and user-friendly
- Automatically finds the correct server address
- Essential for internet communication and website access
- Uses hierarchical distributed database system

**Example:** google.com → IP address → Opens Google website.

# Networking Protocol(TCP)

## TCP Handshake process



# Networking Protocol

## TCP (Transmission Control Protocol)

- Reliable transport layer protocol used for data communication
- Ensures data is delivered accurately and in correct order
- Establishes a connection before data transfer (connection-oriented)
- Uses error checking and retransmission of lost data
- Controls data flow to avoid network congestion
- Provides acknowledgment (ACK) for received data

**Example:** Downloading a file or loading a website in a browser.

# Networking Protocol

## ICMP (Internet Control Message Protocol)

- Network layer protocol used for error reporting and diagnostics
- Helps in checking network connectivity between devices
- Sends error messages when data cannot reach destination
- Used by tools like ping and traceroute
- Does not transfer actual user data
- Works with IP to report network issues
- Helps identify packet loss, delays, or unreachable hosts

**Example:** Using the `ping google.com` command to check if a server is reachable.

# Networking Protocol

## IP (Internet Protocol)

- Core network layer protocol used for addressing and routing data
- Assigns a unique IP address to each device on a network
- Responsible for sending data packets from source to destination
- Works in a connectionless manner (no guaranteed delivery)
- Defines how data is broken into packets and routed
- Supports IPv4 and IPv6 addressing systems
- Works with TCP/UDP for complete communication

**Example:** Sending data from your computer to a website using its IP address (e.g., 192.168.1.1).

# Networking Protocol

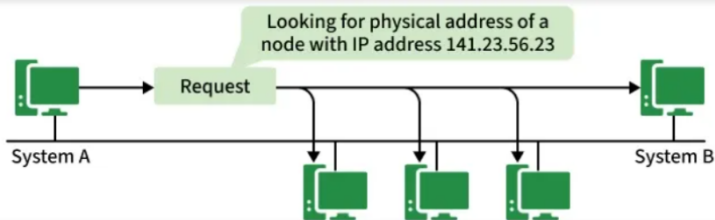
## ARP (Address Resolution Protocol)

- Network layer protocol used to find MAC address from an IP address
- Works within a local network (LAN)
- Maps logical IP address to physical MAC address
- Helps devices communicate on the same network
- Uses ARP request and ARP reply messages
- Stores results in ARP cache for faster future communication
- Essential for data delivery inside LAN

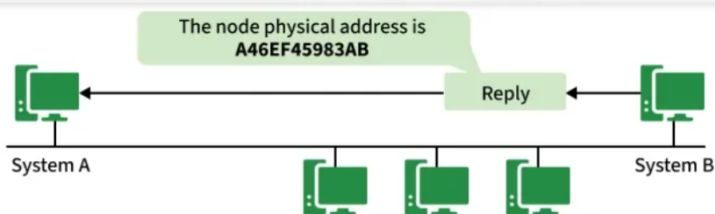
**Example:** A computer finds the MAC address of a router using its IP address to send data.

# ARP

a. ARP request is broadcast



b. ARP reply is unicast



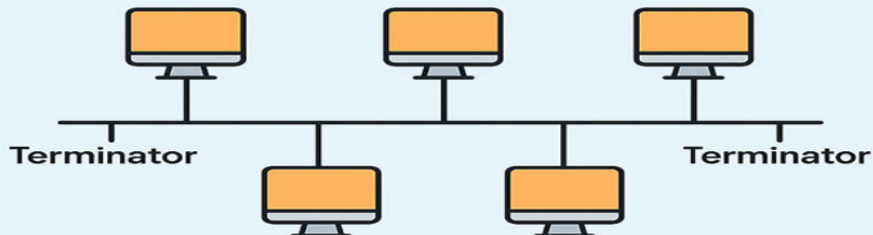
# Network Topology

Network Topology refers to the arrangement and organization of devices in a computer network. It defines how components are structured and how communication is established between them.

- Describes the overall structure of a network
- Helps in network design and planning
- Influences performance and reliability
- Includes Physical Topology: actual layout of devices and cables
- Includes Logical Topology: how data moves between devices

# Bus Topology

## Bus Topology



# Bus Topology

A bus topology is a type of network setup in which all computers and devices are connected to a single main cable called the bus or backbone.

## Features:

- Uses one central cable to connect all devices
- Data travels in both directions along the bus
- Terminators are placed at both ends of the cable to prevent signal reflection

## Advantages:

- Simple and inexpensive to install
- Requires less cable compared to other topologies
- Easy to extend for small networks

# Bus Topology

## Disadvantages:

- If the main cable fails, the entire network stops working
- Performance decreases as more devices are added
- Difficult to troubleshoot faults
- Limited cable length and number of devices

## Applications:

- Small Local Area Networks (LANs)
- Temporary or small office networks
- Early Ethernet networks

## Example:

If five computers are connected to one common cable and share communication through it, that setup is called a bus topology.

# Star Topology

A star topology is a network setup in which all devices are connected to a single central device such as a switch or hub.

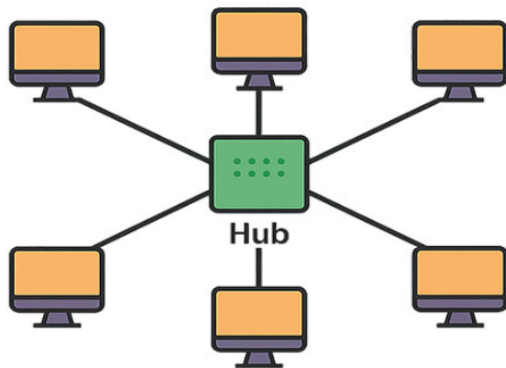
## Features:

- All devices connect to a central hub/switch
- Data passes through the central device before reaching destination
- Each device has a dedicated cable to the hub

## Advantages:

- Easy to install and manage
- Failure of one device does not affect others
- Simple troubleshooting and fault detection
- High performance in modern networks (especially with switches)

## Star Topology



# Star Topology

## Disadvantages:

- If the central hub/switch fails, the whole network stops working
- Requires more cable than bus topology
- Higher installation cost

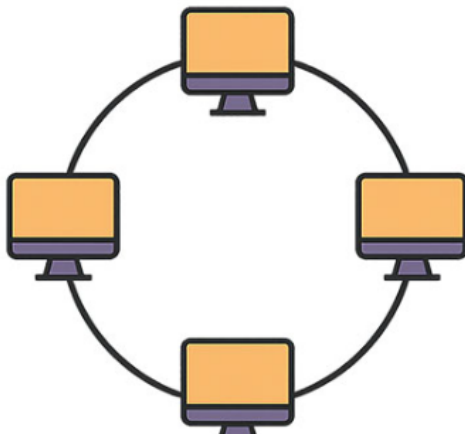
## Applications:

- Home and office LANs
- Schools and computer labs
- Modern Ethernet networks

## Example:

If 10 computers are connected to a single switch, and all communication goes through it, this is a star topology.

# Ring Topology



# Ring Topology

A ring topology is a type of network configuration where each device is connected to exactly two other devices, forming a closed loop (ring shape).

## Features:

- Devices are connected in a circular manner
- Data travels in one direction (uni-directional) or both directions (bi-directional in dual ring)
- Each node acts as a repeater to forward data

## Advantages:

- Equal access to network for all devices
- No central hub required
- Performance is better than bus topology under heavy load

# Ring Topology

## Disadvantages:

- Failure of one device or cable can affect the entire network
- Adding or removing devices can disrupt the network
- Troubleshooting is difficult

## Applications:

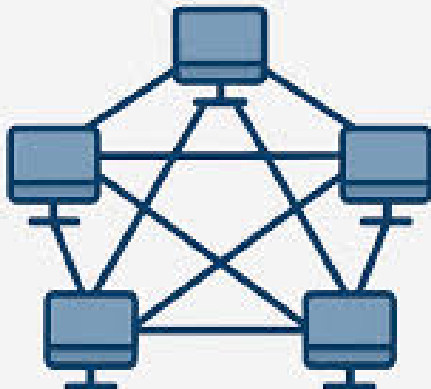
- Token Ring networks (older LAN technology)
- Some metropolitan networks (MANs)
- Fiber optic ring networks in telecom systems

## Example:

If 6 computers are connected in a circle and data moves from one computer to the next until it reaches the destination, that is a ring topology.

# Mesh Topology

## Mesh Topology



# Mesh Topology

A mesh topology is a network structure in which every device is directly connected to every other device, creating multiple paths for data transmission.

## Features:

- Each node is connected to many or all other nodes
- Data can travel through multiple possible paths
- Two types:
  - Full Mesh: every device connects to every other device
  - Partial Mesh: only some devices are fully interconnected

## Advantages:

- Very reliable and fault-tolerant
- If one link fails, data can use another path
- Strong privacy and security (dedicated links)
- No traffic congestion on a single path

# Mesh Topology

## Disadvantages:

- Very expensive (requires lots of cables and ports)
- Difficult to install and maintain
- Complex structure for large networks

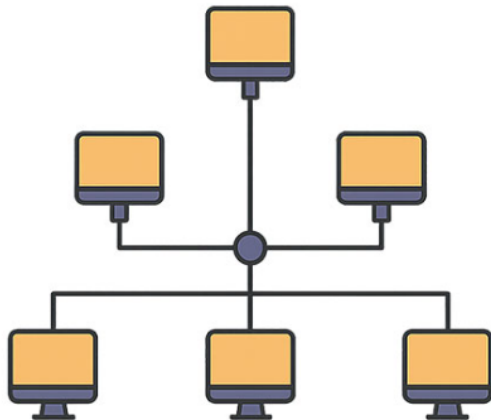
## Applications:

- Military communication systems
- Internet backbone networks
- Critical systems (banking, telecom networks)
- Wireless mesh networks (city-wide Wi-Fi)

## Example:

If 4 computers are each directly connected to all other computers, it forms a mesh topology where multiple communication paths exist between devices.

## Tree Topology



# Tree Topology

A tree topology is a network structure that looks like a hierarchical tree, combining characteristics of both star and bus topologies.

## Features:

- Organized in a hierarchical structure (like a family tree)
- Top node is called the root node
- Below it are multiple levels of branches and sub-branches
- Each branch can have its own star network

## Advantages:

- Easy to expand (scalable)
- Fault isolation is easier than bus topology
- Supports large networks efficiently
- Structured and organized design

# Tree Topology

## Disadvantages:

- If the main backbone (root link) fails, large parts of the network are affected
- More cabling required than simple topologies
- Installation and maintenance can be complex

## Applications:

- Large corporate networks
- University campus networks
- Distributed databases
- Organizational networks with departments and sub-departments

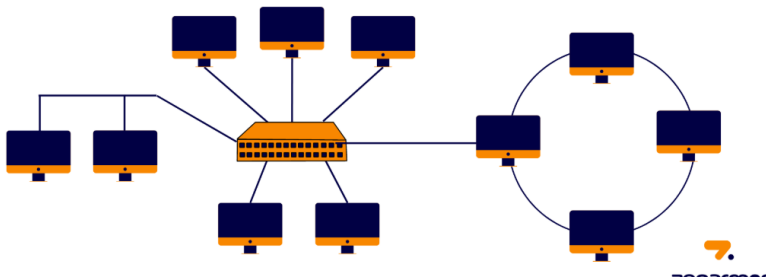
## Example:

A university network where:

- Main server = root node
- Different departments = branches
- Each lab computer cluster = sub-branches

## HYBRID NETWORK TOPOLOGY

A hybrid topology is a type of network topology that combines two or more network topologies, including ring, bus, and mesh topologies



# Hybrid Topology

A hybrid topology is a network structure that combines two or more different topologies (such as star, bus, ring, or mesh) to form a larger and more flexible network.

## Features:

- Mixes multiple topologies in one network
- Each section of the network can use a different topology
- Common in large and complex systems
- Designed to meet specific organizational needs

## Advantages:

- Very flexible and scalable
- Can be designed for high performance and reliability
- Failure in one section may not affect the whole network
- Suitable for large organizations

# Hybrid Topology

## Disadvantages:

- Complex to design and manage
- Expensive due to advanced infrastructure
- Requires skilled maintenance and configuration

## Applications:

- Large corporate networks
- Banking systems
- Universities and campuses
- Internet Service Provider (ISP) networks

## Example:

A company network where:

- Office floors use star topology
- Data center uses mesh topology
- Departments are connected in tree structure

# Information Security(INFOSEC)

Information Security is the process of protecting information from unauthorized access, use, disclosure, modification, or destruction.

- Ensures data stored in computers, networks, and communication systems remains safe and reliable
- Applies to both digital information (files, databases, emails) and physical information (documents, records)
- Important for individuals, organizations, and governments to protect sensitive data
- Helps prevent cyber attacks, data theft, and misuse of information

# Objectives of Information Security

- To protect information from unauthorized users or hackers
- To ensure that data remains confidential and private
- To maintain the accuracy and consistency of data
- To ensure that information is available whenever required
- To protect organizational and personal sensitive data

## Core Principle of Information Security(CIA Triad)



# Confidentiality

- Ensures that information is accessible only to authorized users
- Prevents unauthorized access or disclosure of data
- Protects sensitive information like passwords, personal data, and business secrets

# Integrity

- Ensures that information is accurate, complete, and trustworthy
- Prevents unauthorized modification or tampering of data
- Maintains consistency of information during storage and transmission

# Availability

- Ensures that information and systems are available when needed by authorized users
- Prevents system downtime and service disruption
- Maintains continuous access to data and resources

# Why CIA Triad Alone is Not Enough

The CIA Triad (Confidentiality, Integrity, Availability) is the foundation of Information Security, but it only defines the security goals, not the methods to achieve them.

- CIA Triad defines security objectives, not implementation methods
- Real systems need additional security principles
- Includes Authentication, Authorization, Accounting, and Non-repudiation (AAA + NR)
- These help in proper enforcement of security in real-world systems

# Authentication

AAA + Non-repudiation = Security Mechanisms (How to protect)  
These are used to enforce CIA in real systems.

## Authentication (Who are you?)

- Verifies identity of a user or system
- First step before granting access
- Ensures only valid users enter the system
- Methods: password, OTP, fingerprint, face recognition
- Supports Confidentiality and Integrity
- Without authentication, anyone can access the system

## Authorization (What can you do?)

- Determines user permissions after authentication
- Controls access to resources and actions
- Different roles have different privileges
- Example:
  - Admin → full control
  - User → limited access
- Supports Confidentiality and Integrity
- Prevents misuse of valid accounts

## **Accounting (What did you do?)**

- Records user activities in the system
- Keeps logs like login time, file access, and changes made
- Helps in monitoring and auditing
- Useful for detecting suspicious behavior
- Supports Integrity and system monitoring
- Important for security investigation

## Non-repudiation (Proof of action)

- Ensures a user cannot deny their actions later
- Provides legal and technical proof of transactions
- Uses digital signatures and secure logs
- Example: online banking transaction cannot be denied
- Builds trust and accountability
- Strengthens Integrity in real systems

# Common Security Threats and Vulnerabilities

## Security Threats

- A security threat is any potential danger that can harm a computer system, network, or data
- It can lead to unauthorized access, data loss, or system damage
- Threats can be intentional (hackers) or unintentional (human error)

# Common Security Threats

- **a) Malware (Malicious Software)**

- Malware is harmful software designed to damage or disrupt systems
- Includes viruses, worms, trojans, ransomware, and spyware
- Can delete files, steal data, or take control of a system

- **b) Phishing Attacks**

- Phishing is a social engineering attack where users are tricked into revealing sensitive information
- Fake emails or websites are used to steal passwords, bank details, or personal data
- Example: fake bank login page

# Common Security Threats

- **c) Denial of Service (DoS / DDoS)**

- This attack makes a system or website unavailable to users
- It works by flooding the system with excessive traffic
- Result: system becomes slow or completely crashes

- **d) Man-in-the-Middle (MITM) Attack**

- Attacker secretly intercepts communication between two users
- Can read, modify, or steal data during transmission
- Example: unsafe public Wi-Fi interception

# Common Security Threats

- **e) Password Attacks**

- Attackers try to guess or steal passwords
- Methods include brute force, dictionary attacks, and keylogging
- Weak passwords increase risk of compromise

- **f) SQL Injection (Web Attack)**

- Attackers insert malicious SQL code into input fields of a website
- It allows them to access or modify database data
- Happens when inputs are not properly secured

- **g) Insider Threat**

- Threat caused by people within the organization (employees, staff)
- Can be intentional (malicious insider) or accidental (human error)
- Hard to detect because insiders already have access

# Vulnerabilities

- A vulnerability is a weakness or flaw in a computer system, network, or application
- It can be exploited by attackers to gain unauthorized access or cause damage
- Vulnerabilities increase the risk of security breaches
- They may exist in software, hardware, configuration, or human behavior

# Software Vulnerabilities

- Software vulnerabilities are weaknesses or errors present in computer programs or applications
- They usually occur due to coding mistakes, poor design, or lack of proper testing
- Attackers take advantage of these weaknesses to execute malicious code or disrupt system operations
- If not fixed, they can allow hackers to gain control of the system or steal data

## Examples:

- Unpatched operating systems (not updated with security fixes)
- Bugs or errors in application software
- Buffer overflow issues where extra data overwrites memory

# Hardware Vulnerabilities

- Hardware vulnerabilities are weaknesses found in physical components of a computer system
- These vulnerabilities occur due to faulty design, physical damage, or lack of security controls
- Attackers may exploit physical access to damage or manipulate hardware systems

## **Examples:**

- Faulty or damaged hardware components
- Weak or missing BIOS/firmware security protection
- Unauthorized physical access to servers or computers

# Network Vulnerabilities

- Network vulnerabilities are weaknesses in network design, configuration, or communication systems
- These weaknesses allow attackers to intercept, modify, or disrupt data transmission
- Poor network security increases the risk of cyber attacks

## Examples:

- Open or unused ports left exposed
- Weak firewall rules or improper configuration
- Unsecured Wi-Fi networks without encryption

# Human (User) Vulnerabilities

- Human vulnerabilities are security weaknesses caused by users due to lack of awareness or mistakes
- Humans are often the weakest link in information security
- Attackers use social engineering techniques to exploit users

## **Examples:**

- Using weak or easy-to-guess passwords
- Clicking on phishing emails or malicious links
- Accidentally sharing confidential or sensitive information

# Configuration Vulnerabilities

- Configuration vulnerabilities occur when systems are not properly set up or securely configured
- Incorrect settings create security gaps that attackers can exploit
- These are often caused by poor administration or lack of security knowledge

## Examples:

- Default passwords not changed after installation
- Incorrect firewall or security rule settings
- Giving unnecessary full access to all users

# Operating System Vulnerabilities

- Operating system vulnerabilities are weaknesses found in the OS that can be exploited by attackers
- These vulnerabilities are usually fixed by security updates and patches
- If not updated, systems become highly exposed to attacks

## **Examples:**

- Using outdated operating system versions
- Missing important security updates or patches
- Privilege escalation flaws allowing users to gain admin access

# How Vulnerabilities Are Exploited

- Attackers first scan systems to find weak points or security gaps
- After identifying vulnerabilities, they use threats like malware, phishing, or hacking tools
- Exploitation leads to unauthorized access, data theft, or system damage
- This process is the main method behind most cyber attacks

## **Difference:**

- Vulnerability → Weakness in a system (door left open)
- Threat → Possible danger that tries to enter (burglar)
- Attack → Actual action of breaking in (stealing or damaging)

# Security Best Practices for Server and Network Security

## Security best practices

Security best practices are the recommended guidelines used to protect servers, networks, and data from cyber threats.

- They help reduce vulnerabilities and improve overall system security
- Ensure confidentiality, integrity, and availability of information systems
- Provide structured methods to prevent cyber attacks

# Security Best Practices for Servers

## ● 1. Regular Software Updates and Patching

- Servers must be regularly updated with the latest security patches
- Updates fix known vulnerabilities in the operating system and applications
- Outdated software is a major target for attackers

## ● 2. Strong Authentication Mechanism

- Use strong passwords that include letters, numbers, and symbols
- Enable multi-factor authentication (MFA) for extra security
- Prevents unauthorized access to the server

# Security Best Practices for Servers

## ● 3. Access Control and User Management

- Only authorized users should have server access
- Follow the principle of least privilege (minimum access required)
- Remove unused or inactive user accounts

## ● 4. Firewall Configuration

- Configure firewall to allow only necessary traffic
- Block all unnecessary ports and services
- Helps prevent unauthorized access and attacks

# Security Best Practices for Servers

## ● 5. Data Backup and Recovery

- Regular backups of important server data should be maintained
- Backups should be stored in a secure location
- Helps recover data after system failure or cyber attack

## ● 6. Antivirus and Anti-malware Protection

- Install updated antivirus software on servers
- Regularly scan for malware and suspicious activities
- Helps prevent system infection

## ● 7. Server Hardening

- Disable unnecessary services and applications
- Secure configuration of operating system settings
- Reduces attack surface of the server

# Security Best Practices for Networks

## ● 1. Secure Network Architecture

- Design network with proper segmentation (separating internal and external networks)
- Use DMZ (Demilitarized Zone) for public services
- Limits damage in case of attack

## ● 2. Use of Firewalls and Intrusion Detection Systems (IDS/IPS)

- Firewalls monitor and control incoming and outgoing traffic
- IDS/IPS detect and prevent suspicious network activities
- Provides real-time protection against attacks

# Security Best Practices for Networks

## ● 3. Encryption of Data Transmission

- Use encryption protocols like HTTPS, SSL/TLS, and VPN
- Protects data while it is being transmitted over networks
- Prevents interception by attackers

## ● 4. Secure Wi-Fi Networks

- Use WPA2/WPA3 encryption for wireless networks
- Avoid open or public Wi-Fi for sensitive operations
- Change default router passwords

# Security Best Practices for Networks

## ● 5. Network Monitoring and Logging

- Continuously monitor network traffic for unusual activity
- Maintain logs of network access and events
- Helps in detecting and investigating attacks

## ● 6. Regular Vulnerability Assessment

- Conduct security audits and penetration testing
- Identify and fix weak points in the network
- Improves overall network security

## ● 7. Access Control in Network Devices

- Restrict access to routers, switches, and servers
- Use secure protocols like SSH instead of Telnet
- Disable unused network services

# Fundamentals of Network Reference Models

A Network Reference Model is a conceptual framework used to understand how data is transmitted between computers in a network. It divides communication tasks into layers so that each layer has a specific role.

The most important reference model is the OSI Model developed by the International Organization for Standardization.

## Functions of Network Reference Model

- Standardizes network communication
- Ensures compatibility between devices
- Simplifies network design
- Helps in troubleshooting network problems
- Supports protocol development

# Types of Network Reference Models

## 1. OSI Model (7 Layers)

- Theoretical model
- Used for learning and understanding networks
- Clearly defines 7 layers
- Layers: Application → Presentation → Session → Transport → Network → Data Link → Physical

## 2. TCP/IP Model (4 Layers)

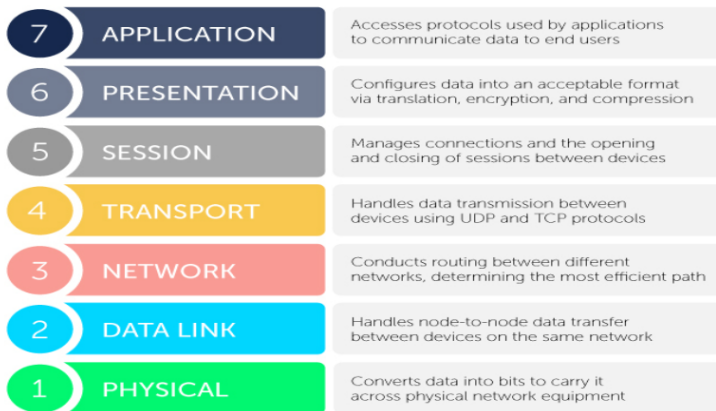
- Practical model used in real internet systems
- Developed for real communication networks
- Layers: Application → Transport → Internet → Network Access

# OSI Model

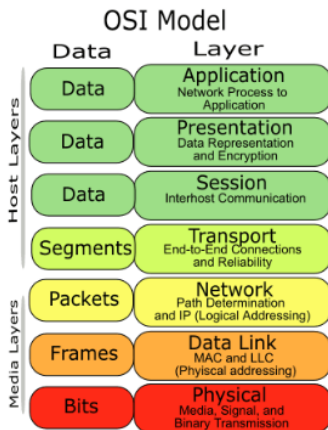
The OSI Model is a conceptual framework created by the International Organization for Standardization (ISO) to describe how data is transmitted across a network using a structured seven-layer architecture.

- Divides network communication into seven functional layers
- Assigns specific responsibilities to each layer
- Promotes compatibility between different networking systems
- Simplifies network design, implementation, and troubleshooting

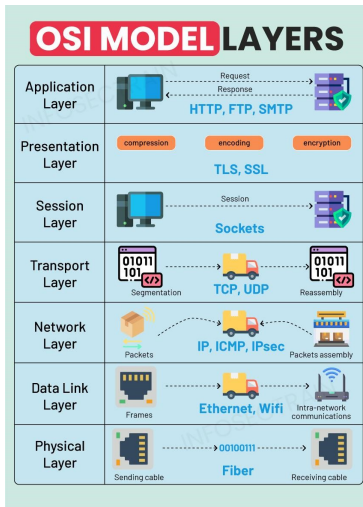
## The seven layers of the OSI model



# OSI model



# OSI model

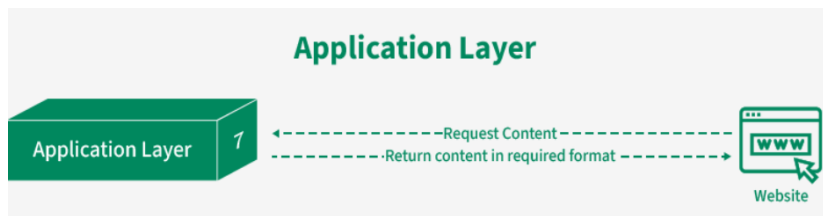


## Layer 7: Application Layer

The Application Layer is the top layer of the OSI model and connects users with the network.

- Helps software like browsers and email apps send and receive data over the internet
- Uses protocols like HTTP/HTTPS, SMTP, FTP, and DNS for communication
- Allows users to access remote computers and systems as if they are local
- Supports file transfer, file management, and access to remote resources
- Provides services to find and manage network information easily

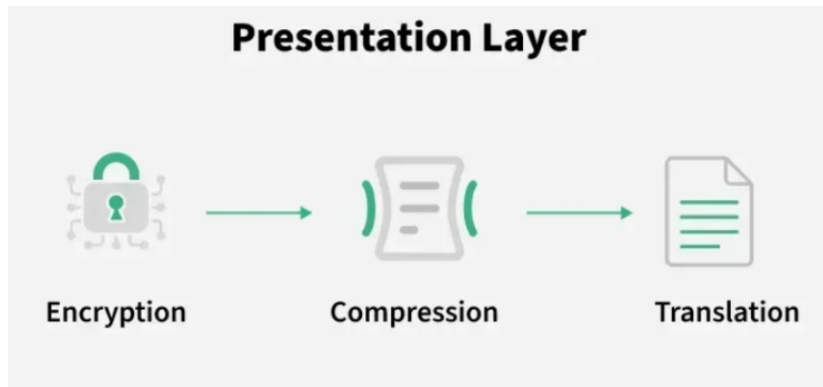
# Application Layer



## Layer 6: Presentation Layer

Often called the **Translation Layer**, the Presentation Layer ensures that data is formatted, secured, and compressed so that the receiving application can correctly interpret it.

- **Translation:** Converts data into a standardized format for network transmission, bridging differences between systems
- **Standards & Formats:** Handles encoding of media using formats such as JPEG, MPEG, and GIF
- **Encryption/Decryption:** Converts plaintext into ciphertext and back using protocols like SSL/TLS
- **Compression:** Reduces data size to improve transmission efficiency and speed



## Layer 5: Session Layer

The Session Layer acts as the "dialogue manager," governing the opening, closing, and security of communication channels between two devices.

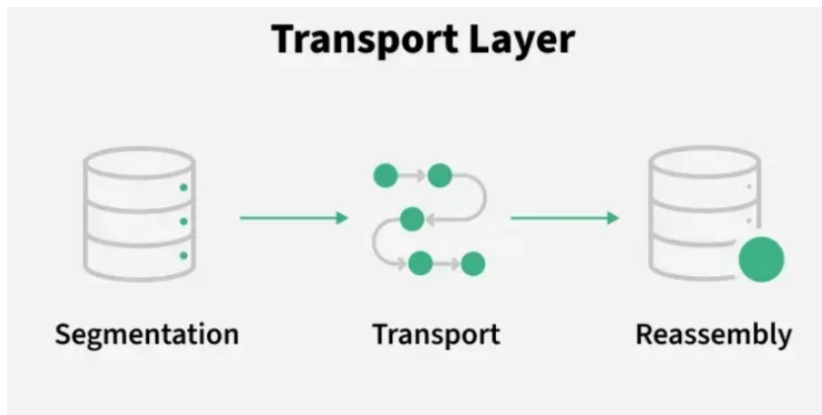
- **Session Lifecycle:** Manages establishment, maintenance, and termination of connections between applications
- **Authentication & Security:** Ensures communicating parties are verified and the connection is secure
- **Synchronization:** Adds checkpoints to data streams to resume transfers after failure
- **Dialog Control:** Manages half-duplex (alternating) and full-duplex (simultaneous) communication
- **Example:** Maintains active sessions in web-based messaging applications

## Layer 4: Transport Layer

The Transport Layer ensures end-to-end delivery of entire messages. It acts as a liaison between the Application Layer and the Network Layer.

- **Data Unit:** Data is broken into segments
- **Service Point Addressing:** Uses port numbers (e.g., Port 80 for web traffic) to deliver data to the correct application
- **Segmentation & Reassembly:** Splits large messages into smaller segments and reassembles them at the destination
- **Protocols:** TCP (reliable), UDP (fast)
- **Connection-Oriented (TCP):** Uses handshake, ensures reliability with acknowledgements and error checking
- **Connectionless (UDP):** Faster transmission without guaranteed delivery

# Transport layer

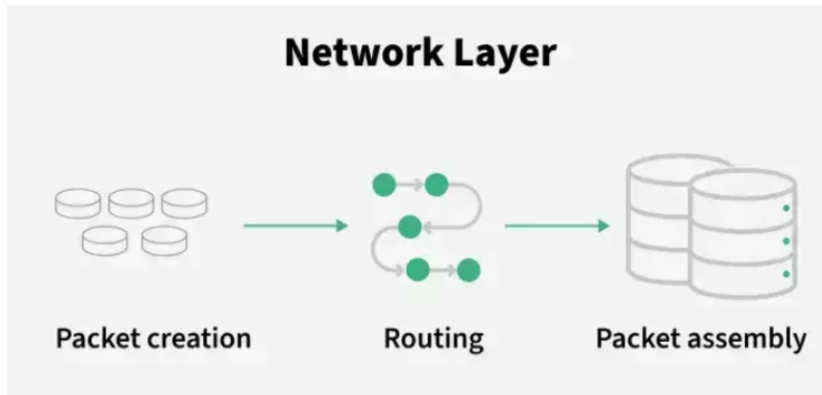


## Layer 3: Network Layer

The Network Layer manages data transmission between hosts across different networks by handling logical addressing and path finding.

- **Data Unit:** Data segments are encapsulated into packets
- **Logical Addressing:** Assigns unique IP addresses (sender and receiver) in the packet header
- **Routing:** Determines the most efficient path from source to destination across networks
- **Hardware:** Primarily implemented via routers
- **Inter-networking:** Enables communication between different networks by directing traffic correctly

# Network layer



## Layer 2: Data Link Layer (DLL)

The Data Link Layer serves as the bridge between the physical hardware and the logical network, ensuring reliable node-to-node delivery of data. Its main goal is error-free transfer across the physical medium.

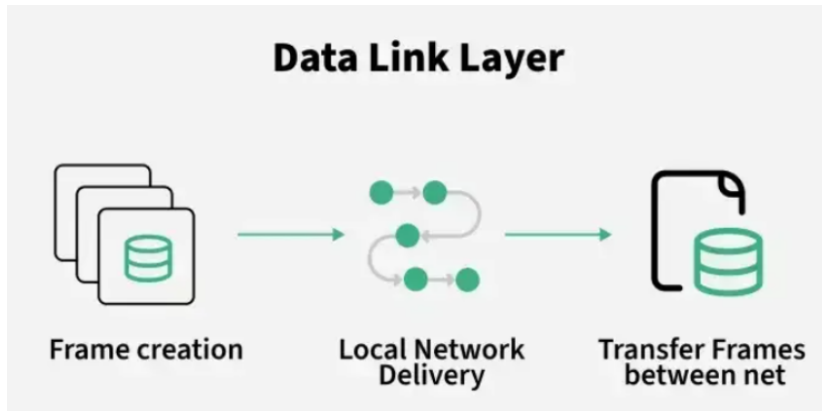
- **Data Packaging (Framing):** Divides packets into frames by adding start and stop bits for proper data recognition
- **Addressing and Hardware:** Uses MAC (physical) addresses in frame headers to identify sender and receiver
- Common devices: switches and bridges

## Layer 2: Data Link Layer (DLL)

- **Sublayers:**

- **LLC (Logical Link Control):** Manages flow and error control and acts as an interface for upper-layer protocols
- **MAC (Media Access Control):** Determines how devices access the transmission medium and handles hardware addressing
- **Error Control:** Detects and retransmits damaged or lost frames to maintain data integrity
- **Flow Control:** Synchronizes data rate between sender and receiver to prevent bottlenecks or data loss
- **Access Control:** Manages which device can transmit on a shared medium to avoid collisions

# Data Link Layer



# Layer 1: Physical Layer

The Physical Layer is the foundation of the OSI model, acting as the bridge for actual physical connections between devices. Its primary mission is the transmission of raw, unstructured bitstreams over a physical medium from one node to the next.

- **Core Responsibility:** Manages hardware-level transmission of individual bits (0s and 1s)
- **Signal Conversion:** Converts digital data into physical signals and back again
- **Essential Hardware:** Hubs, repeaters, modems, cables
- **Bit Synchronization:** Ensures sender and receiver are synchronized using timing signals
- **Bit Rate Control:** Defines transmission speed (bits per second)
- **Physical Topologies:** Supports bus, star, and mesh layouts
- **Transmission Modes:** Simplex, half-duplex, full-duplex communication

# Physical Layer

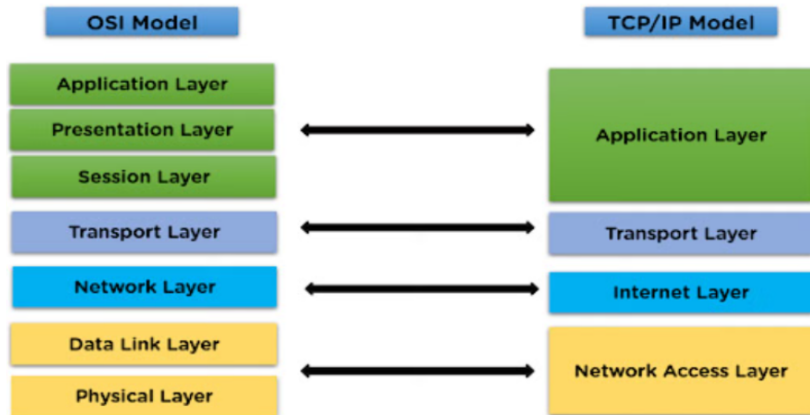


# TCP/IP Model

The TCP/IP model (Transmission Control Protocol / Internet Protocol model) is a 4-layer networking model that explains how data is transmitted between devices over the Internet or any network.

- It is a conceptual model for understanding networking
- It is also a practical protocol suite used in real communication systems
- Defines how data moves from one device to another over networks

# TCP/IP model



## 4.Application Layer

The top layer of the TCP/IP model, closest to the user, where applications like web browsers, email clients, and file-sharing tools interact with the network. It provides an interface between user software and the lower layers that handle data transmission, enabling seamless communication over the network.

- Acts as a bridge between user applications and lower network layers
- Supports protocols such as HTTP, FTP, SMTP, and DNS
- Handles data formatting so information is correctly understood by sender and receiver
- Provides encryption for secure communication
- Manages sessions to track ongoing connections

### 3. Transport Layer

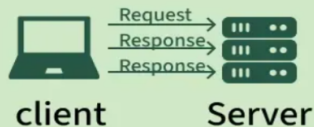
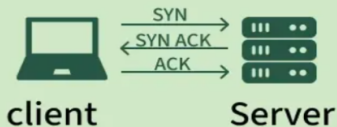
Ensures reliable and efficient delivery of data between devices, managing segmentation, ordering, and retransmission as needed.

- **Segmentation and Reassembly:** Breaks large messages into packets and reassembles them at the destination
- **Reliable Delivery & Error Handling:** TCP checks for errors, resends lost data, and ensures correct order
- **Fast Communication:** UDP provides low-latency, connectionless delivery without error checking
- **Flow Control:** Prevents the receiver from being overwhelmed by regulating data flow
- **Multiplexing:** Uses port numbers to allow multiple applications to share the network simultaneously

# Transport Layer

## Transport Layer Protocols

TCP ——— VS ——— UDP



# Transport layer

**i.TCP (Transmission Control Protocol)** TCP is used when reliability and accuracy are important. It ensures that data is delivered exactly as sent.

- **Error Checking:** Uses checksums to detect errors and ensure data integrity
- **Retransmission:** Automatically resends lost or corrupted data
- **Ordered Delivery:** Ensures packets arrive in the correct sequence
- **Connection-Oriented:** Establishes a connection before data transfer and maintains a stable session
- **Use Cases:** Web browsing, file downloads, email communication, and other reliable applications

## ii.UDP (User Datagram Protocol)

UDP is used when speed is more important than perfect accuracy. It is faster but does not guarantee reliable delivery.

- **No Error Checking:** UDP does not verify whether the data has errors
- **No Retransmission:** Lost or damaged data is not resent
- **No Ordered Delivery:** Packets may arrive out of order and are not corrected
- **Lightweight:** Faster and uses fewer system resources due to minimal overhead
- **Use Cases:** Live video streaming, online gaming, VoIP calls, and real-time applications

## 2. Internet Layer

Responsible for addressing, packaging, and routing data packets so they can travel across networks and reach the correct destination device. It ensures that data can move between different networks efficiently.

- **Logical Addressing:** Assigns IP addresses to identify source and destination devices
- **Packet Routing:** Determines the best path for data to travel across networks
- **Fragmentation and Reassembly:** Breaks large packets into smaller ones and rebuilds them at the destination
- **Protocol Support:** Uses IP (Internet Protocol), ICMP for error reporting, and ARP for address resolution

# 1. Network Access (Link Layer)

Responsible for physically transmitting data over network hardware, including cables, switches, and wireless connections. It handles how data is formatted for the network medium and ensures it reaches the next device on the path.

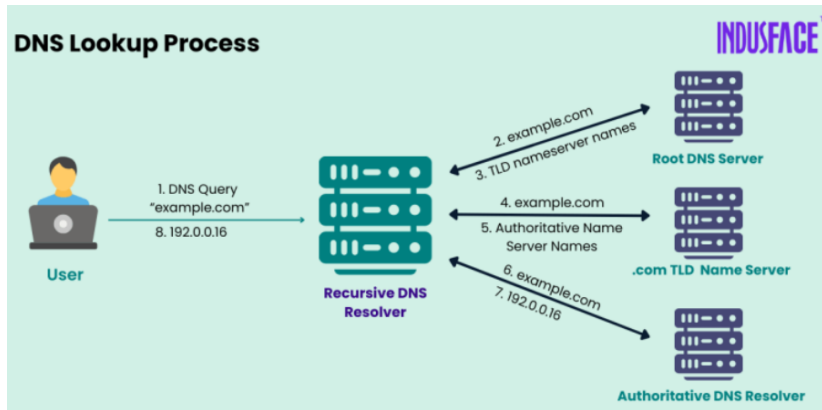
- **Physical Transmission:** Sends and receives raw bits over Ethernet, fiber optics, or Wi-Fi
- **Framing:** Organizes data into frames for proper transmission and recognition by devices
- **Error Detection:** Detects transmission errors using checksums or CRC
- **MAC Addressing:** Uses hardware addresses to identify devices within the same network segment
- **Access Control:** Manages how devices share the physical medium and avoids collisions

# DNS (Domain Name System)

DNS (Domain Name System) converts domain names into IP addresses so computers can communicate over the internet.

- Helps users access websites using easy names instead of numbers
- Works like the internet's phonebook
- Makes web browsing simple and user-friendly
- Example:
  - `www.example.com` → `93.184.216.34`

# DNS working process



# Working of DNS (Step 1–2)

## Step 1: User Request

- User enters a URL in the browser
- Example: `www.example.com`
- Browser first checks:
  - Browser cache
  - Operating system DNS cache

## Step 2: DNS Resolver

- If not found locally, request goes to a DNS Resolver
- Usually provided by ISP or public DNS like Google DNS
- Resolver acts as a middleman to find the IP address

# Working of DNS (Step 3–5)

## Step 3: Root DNS Server

- Resolver asks the Root DNS Server
- Root server directs request to the correct TLD server
- Example: “.com” → .com TLD server

## Step 4: TLD Server

- Handles domains like .com, .org, .net, .edu
- Directs request to the authoritative server

## Step 5: Authoritative DNS Server

- Stores actual IP address and domain records
- Example: www.example.com → 93.184.216.34

# Working of DNS (Step 6–7)

## Step 6: Response Returned

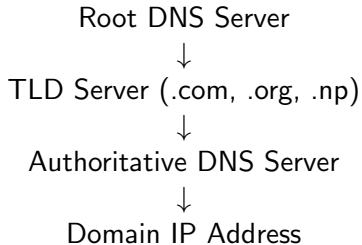
- IP address is returned back to the browser
- Path:
  - Authoritative Server → Resolver → Browser

## Step 7: Website Loads

- Browser uses the IP address to connect to the web server
- Requested website is displayed to the user

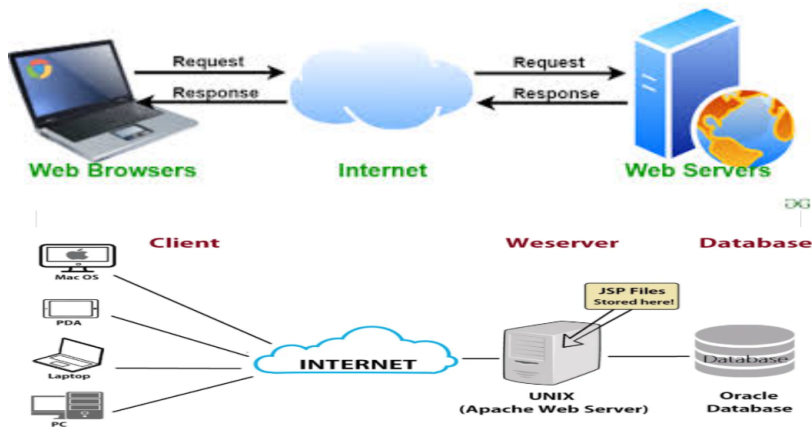
# DNS Hierarchy Structure

DNS is organized like a tree structure:



This hierarchical structure makes DNS scalable and fast worldwide.

# Web Server



A web server is a system (hardware + software) that stores, processes, and delivers web content to users over a network using HTTP/HTTPS.

## Main Function

- Accept requests from clients (browsers)
- Process those requests
- Send back web pages (responses)

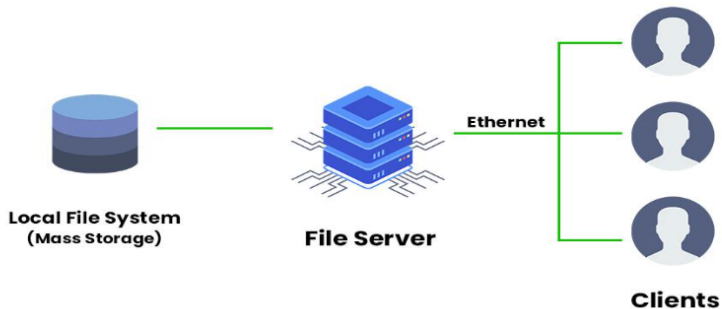
# Working of Web Server (Step-by-Step)

- User opens a web browser (Chrome, Firefox, etc.)
- User types a URL (e.g., [www.google.com](http://www.google.com))
- Browser sends an HTTP/HTTPS request
- Web server receives the request
- Server processes the request (may access database or files)
- Server sends back response (HTML, CSS, JS files)
- Browser displays the webpage

# Features of Web Server

- Hosts websites
- Handles multiple users simultaneously
- Provides security (HTTPS, authentication)
- Supports dynamic applications
- Works 24/7 online

# File Server



# File Server

A file server is a computer system (hardware + software) that stores files and allows multiple users over a network to access, upload, download, and manage those files.

## Main Function

- Store files centrally
- Allow network users to access shared files
- Manage file permissions (read, write, delete)
- Provide data backup and security

# Working of File Server

- User connects to the network (LAN or Internet)
- User sends a request for a file
- File server checks user permission
- If allowed, server locates the file
- Server sends file over the network
- User downloads or opens the file

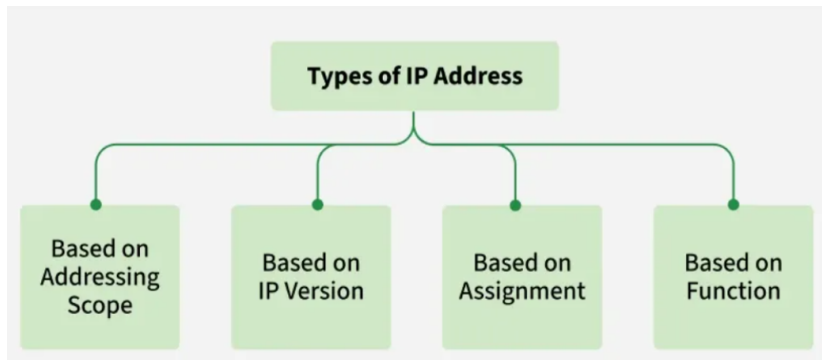
# IP Address (Internet Protocol Address)

An IP Address (Internet Protocol Address) is a unique numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.

It serves two main purposes:

- Identifying a device on the network
- Locating the device to enable communication with other devices over a network like the Internet

# IP address



# Type of IP::Based on Addressing Scope

## Private IP Addresses

- Used inside private networks (home, office, school LAN)
- Not routable on the internet (cannot be accessed directly from outside)
- Devices with private IPs need a router with NAT (Network Address Translation) to access the internet
- Used for internal communication between devices in the same network
- Private IPs must be unique only within the local network, not globally

## IPv4 Private Address Ranges

- 10.0.0.0 – 10.255.255.255 (Class A)
- 172.16.0.0 – 172.31.255.255 (Class B)
- 192.168.0.0 – 192.168.255.255 (Class C)

# Type of IP::Based on Addressing Scope

## Public IP Addresses

- Assigned to devices that directly access the internet
- Unique across the entire internet (no duplication globally)
- Used for global communication between devices over the internet
- Assigned by Internet Service Providers (ISPs)

## Characteristics

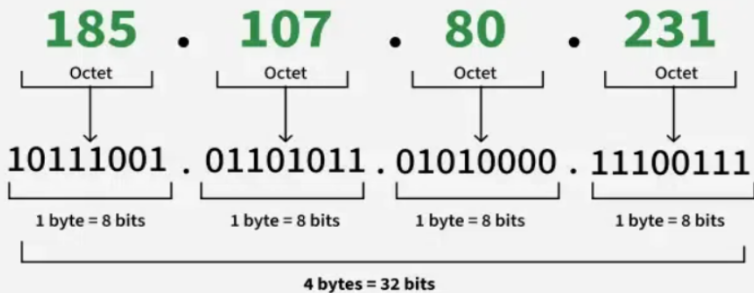
- **Uniqueness:** Each public IP is globally unique
- **Accessibility:** Can be accessed from anywhere on the internet

# Type of IP:Based on IP Version

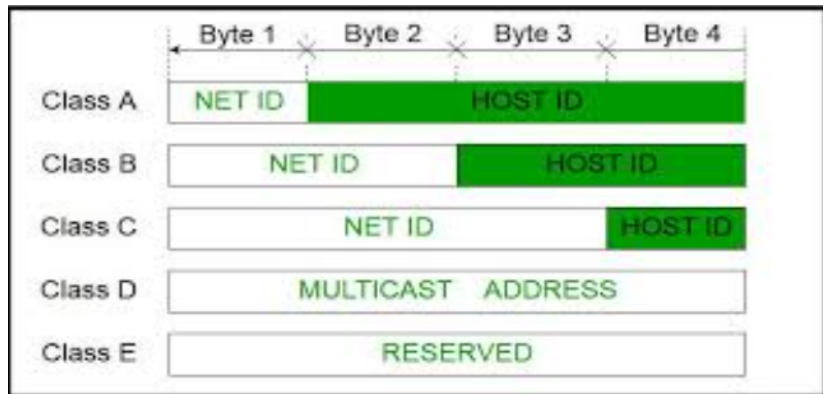
## IPv4 (Internet Protocol Version 4)

- Most commonly used IP address format
- Written as four numbers (octets) separated by dots (e.g., 192.168.1.1)
- Each IP address has 4 octets = 32 bits total
- Each octet is 8 bits (1 byte)
- Each octet value ranges from 0 to 255 ( $2^8 = 256$  values)
- Provides about 4.3 billion unique addresses
- Widely used but has a limited address space

## IPv4 Address Format



# IPv4



# Type of IP:Based on IP Version

## IPv6 (Internet Protocol Version 6)

- Developed to overcome the shortage of IPv4 addresses
- Uses a 128-bit address format (much larger address space than IPv4)
- Written as 8 groups of hexadecimal numbers
- Each group contains 4 hexadecimal digits
- Groups are separated by colons (:) )

## IPv6 address

**2001: 0DC8: E004:0001:0000:0000:0000: F00A**

16 bits: 16 bits: 16 bits: 16 bits: 16 bits: 16 bits: 16 bits: 16 bits

**128 Bits**

# Type of IP: Based on Assignment

## Static IP Addresses

- Static IP Addresses are permanently assigned to a device
- Typically used for servers or devices that need a constant address
- Reliable for network services that require regular access such as websites and remote management

# Type of IP:Based on Assignment

## Dynamic IP Addresses

- Temporarily assigned from a pool of available addresses by the Dynamic Host Configuration Protocol (DHCP)
- Cost-effective and efficient for providers
- Ideal for consumer devices that do not require permanent addresses

# Type of IP:: Based on Function

## Unicast Address

- Data is sent from one sender to one specific receiver
- Uses a unique IP address for communication
- Most common type of network communication
- Purpose: One-to-one communication

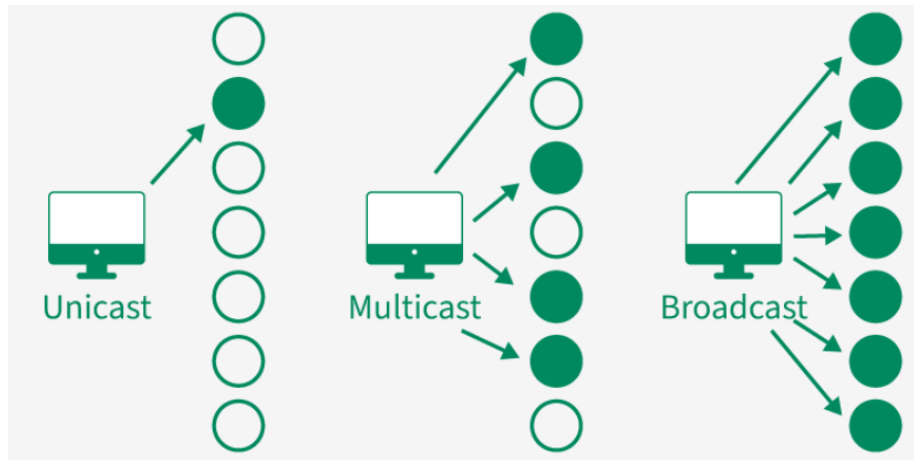
## Multicast Address

- Data is sent from one source to multiple selected receivers
- Only devices in the multicast group receive the data
- More efficient than broadcasting
- Purpose: One-to-many (selected group) communication

# Type of IP:: Based on Function

## Broadcast Address

- Data is sent from one sender to all devices in the network
- All devices in the local network receive the message automatically
- Used for network-wide announcements and discovery
- Purpose: One-to-all communication



# Classes of IPv4 Address

There are around 4.3 billion IPv4 addresses. Managing all those addresses without any classification is difficult.

For easier management and assignment, IP addresses are organized in numeric order and divided into the following 5 classes:

# Classes of IPv4 Address

There are around 4.3 billion IPv4 addresses. Managing all those addresses without any classification is difficult.

For easier management and assignment, IP addresses are organized in numeric order and divided into the following 5 classes:

# Classes of IPv4 Address

- **Class A (1.0.0.0 to 127.255.255.255):** Used for very large networks (e.g., multinational companies). Supports up to 16 million hosts per network.
- **Class B (128.0.0.0 to 191.255.255.255):** Used for medium-sized networks, such as large organizations. Supports up to 65,000 hosts per network.
- **Class C (192.0.0.0 to 223.255.255.255):** Used for smaller networks, like small businesses or home networks. Supports up to 254 hosts per network.
- **Class D (224.0.0.0 to 239.255.255.255):** Reserved for multicast groups (sending data to multiple devices). Not used for normal networking.
- **Class E (240.0.0.0 to 255.255.255.255):** Reserved for experimental and future use.

# classful address

Five Different Classes of IPv4 Addresses

| Class                     | First Octet decimal (range) | First Octet binary (range) | IP range                  | Subnet Mask   | Hosts per Network ID | # of networks |
|---------------------------|-----------------------------|----------------------------|---------------------------|---------------|----------------------|---------------|
| Class A                   | 0 — 127                     | 0XXXXXXXX                  | 0.0.0.0-127.255.255.255   | 255.0.0.0     | $2^{24}-2$           | $2^7$         |
| Class B                   | 128 — 191                   | 10XXXXXXXX                 | 128.0.0.0-191.255.255.255 | 255.255.0.0   | $2^{16}-2$           | $2^{14}$      |
| Class C                   | 192 — 223                   | 110XXXXXX                  | 192.0.0.0-223.255.255.255 | 255.255.255.0 | $2^8-2$              | $2^{21}$      |
| Class D<br>(Multicast)    | 224 — 239                   | 1110XXXX                   | 224.0.0.0-239.255.255.255 |               |                      |               |
| Class E<br>(Experimental) | 240 — 255                   | 1111XXXX                   | 240.0.0.0-255.255.255.255 |               |                      |               |

# Why Classless Addressing (CIDR)?

Classless addressing (CIDR) is used because the old class-based system (Class A, B, C) is inefficient and wastes a lot of IP addresses.

## Why Classless Addressing?

- **Efficient use of IP addresses**
  - In classful system, networks come in fixed sizes (A, B, C)
  - Many IPs get wasted (especially Class B)
- **Flexible subnet sizes**
  - You can choose any prefix like /25, /26, /30, etc.
  - Network size is not fixed anymore
- **Better support for subnetting**
  - Large networks can be divided into smaller subnets easily
- **Reduces IP address exhaustion**
  - IPv4 addresses are limited ( 4.3 billion)
  - CIDR helps extend usability of IPv4
- **Improves routing efficiency**

## IP Addressing: 192.168.10.0/25

- Find the subnet mask
- Find the total number of networks that can be created
- Find the total number of hosts on each network
- Find the network ID and broadcast ID of each network

# Subnet Mask

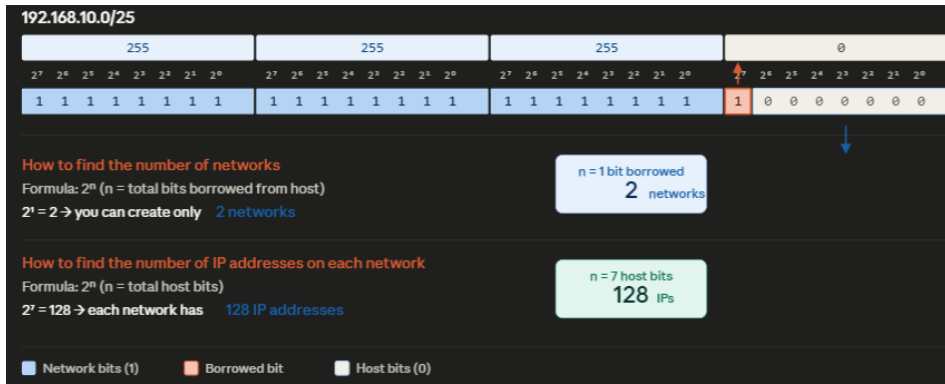
**IP Address:** 192.168.10.0/25

**Class:** Class C (192.0.0.0 – 223.255.255.255)

**Subnet Mask Calculation:**

- 11111111.11111111.11111111.10000000 (1 bit borrowed from host ID)
- $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$
- 255.255.255.128

# IP Addressing: 192.168.10.0/25



# IP Addressing: 192.168.10.0/25

192.168.10.0/25

| 255            |                |                |                |                |                |                |                | 255            |                |                |                |                |                |                |                | 255            |                |                |                |                |                |                |                | 0              |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> | 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> | 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> | 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> |
| 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |

**How to find the number of networks**

$2^n$  (n = total bits borrowed from host)

$2^1 = 2$  (you can create only two networks)

**How to find the number of IP addresses on each network**

$2^n$  (n = total host bits)

$2^7 = 128$  (each network will have 128 IP addresses)

**How to find the number of hosts in each network**

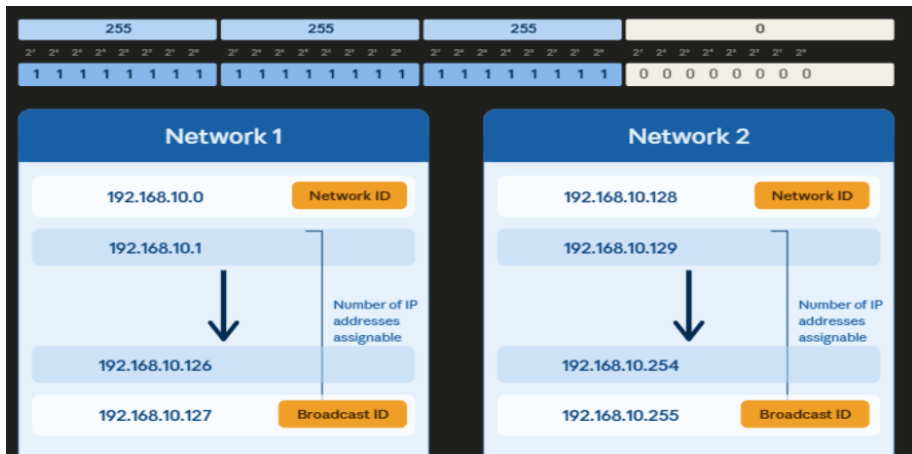
$2^n - 2$  (n = total remaining host bits)

$2^7 - 2 = 126$  (126 host IP addresses on each network)

**Note:** In every network, the first IP address is reserved for the network ID and the last IP address is reserved for the broadcast ID.

■ Network bits (1)    ■ Borrowed bit    ■ Host bits (0)

# IP Addressing: 192.168.10.0/25



## IP Addressing: 192.168.10.0/26

- Find the subnet mask
- Find the total number of networks that can be created
- Find the total number of hosts on each network
- Find the network ID and broadcast ID of each network

# Subnet Mask (/26)

**IP Class:** Class C (192.0.0.0 – 223.255.255.255)

- 11111111.11111111.11111111.11000000 (2 bits borrowed from host ID)
- $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$
- 255.255.255.192

# Subnetting Calculations

## How to find the number of networks

- $2^n$  ( $n$  = total bits borrowed from host)
- $2^2 = 4$  (you can create only 4 networks)

## How to find the number of IP addresses on each network

- $2^n$  ( $n$  = total host bits)
- $2^6 = 64$  (each network will have 64 IP addresses)

## How to find the number of hosts in each network

- $2^n - 2$  ( $n$  = total remaining host bits)
- $2^6 - 2 = 64 - 2 = 62$  (62 usable host IP addresses)

## Subnetting Result: 192.168.10.0/26

- Total No. of Networks: 4
- Total No. of IP Addresses on each network: 64
- Total No. of Hosts on each network:  $64 - 2 = 62$

### Network IDs

- Network 1: 192.168.10.0
- Network 2: 192.168.10.64 (0 + 64)
- Network 3: 192.168.10.128 (64 + 64)
- Network 4: 192.168.10.192 (128 + 64)

# Network ID and Broadcast ID

## 1st Network

- Network ID: 192.168.10.0
- Broadcast ID: 192.168.10.63

## 2nd Network

- Network ID: 192.168.10.64
- Broadcast ID: 192.168.10.127

## 3rd Network

- Network ID: 192.168.10.128
- Broadcast ID: 192.168.10.191

## 4th Network

- Network ID: 192.168.10.192
- Broadcast ID: 192.168.10.255

# Subnetting numerical

**Given IP Address:**

172.168.10.0/22

**Find the following:**

- Subnet Mask
- Total number of networks that can be created
- Total number of hosts on each network
- Network ID and Broadcast ID of each network

# Subnetting Example: 172.168.10.0/22

**IP Address:** 172.168.10.0/22

## **Class Identification**

172.x.x.x lies in **Class B** IP address range.

**Default subnet mask for Class B:**

/16

**Given subnet:**

/22

# 1. Find the Subnet Mask

CIDR /22 means:

11111111.11111111.11111100.00000000

**Subnet Mask:**

255.255.252.0

## 2. Total Number of Networks

**Borrowed bits:**

$$22 - 16 = 6$$

**Total networks:**

$$2^6 = 64$$

**Answer:**

64 networks

### 3. Total Hosts on Each Network

**Host bits:**

$$32 - 22 = 10$$

**Total hosts:**

$$2^{10} = 1024$$

**Usable hosts:**

$$1024 - 2 = 1022$$

(2 addresses reserved for Network ID and Broadcast ID)

**Answer:**

1022 hosts per network

## 4. Network ID and Broadcast ID

For /22:

**Block size in 3rd octet:**

$$256 - 252 = 4$$

**Subnets increase by 4:**

$$0 - 3, 4 - 7, 8 - 11, 12 - 15, \dots$$

Given IP:

$$172.168.10.0$$

10 falls in range:

$$8 - 11$$

**Network ID:**

# Subnetting

## 172.168.10.0/22

| 255            |                |                |                |                |                |                |                | 255            |                |                |                |                |                |                |                | 0              |                |                |                |                |                |                |                | 0              |                |                |                |                |                |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> | 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> | 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> | 2 <sup>7</sup> | 2 <sup>6</sup> | 2 <sup>5</sup> | 2 <sup>4</sup> | 2 <sup>3</sup> | 2 <sup>2</sup> | 2 <sup>1</sup> | 2 <sup>0</sup> |
| 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |

**Total Network:** 64

**Host on each network:** 1022

**How to find the network ID of each network**

Just find the value of last octet borrowed from the host

$$2^2 = 4$$

| Network Number | Network ID             | Host ID                         | Broadcast ID    |
|----------------|------------------------|---------------------------------|-----------------|
| 1              | 172.168. <b>0</b> .0   | 172.168.0.1 - 172.168.3.254     | 172.168.3.255   |
| 2              | 172.168. <b>4</b> .0   | 172.168.4.1 - 172.168.7.254     | 172.168.7.255   |
| 3              | 172.168. <b>8</b> .0   | 172.168.11.1 - 172.168.11.254   | 172.168.11.255  |
| 4              | 172.168. <b>12</b> .0  | 172.168.12.1 - 172.168.15.254   | 172.168.15.255  |
| 5              | 172.168. <b>16</b> .0  | 172.168.16.1 - 172.168.19.254   | 172.168.19.255  |
|                |                        |                                 |                 |
| 64             | 172.168. <b>252</b> .0 | 172.168.252.1 - 172.168.255.254 | 172.168.255.255 |

# Final Answers

| Item              | Answer         |
|-------------------|----------------|
| Subnet Mask       | 255.255.252.0  |
| Total Networks    | 64             |
| Hosts per Network | 1022           |
| Network ID        | 172.168.8.0    |
| Broadcast ID      | 172.168.11.255 |